

>> SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL, HTML, CSS, WebAssembly, Bash

Frameworks & Libraries: OpenCV, Numpy, Scipy, Pandas, Tensorflow / Keras, PIL, Django, Jinja, Apollo GraphQL

Tools & Technologies: Git, Google Cloud Platform (GCP), Google Colab, Jupyter Notebook, SageMath

>> PROJECTS

Colonel By Online Judge Programming Platform :: cboj.ca

July 2020 - Present

- Extended and open-sourced a platform hosted with **Google Cloud Platform (GCP)** for a high school computer science curriculum's assessments using **Python**, **Django**, **Jinja**, **HTML** and **CSS** — reaching **90+ daily users** in 2021
- Redesigned additional privacy levels with custom point-tallying using **Python** and **MySQL** for student assessments
- Monitored server with **Bash** to meet teachers' specifications, achieving **100+ days** of continuous uptime
- Integrated a user-friendly portal for teachers to upload customized verification scripts

3D Chess

May - June 2021

- Built a three-dimensional variant of chess using **JavaScript**, **WebAssembly** and **C++**
- Designed an autonomous player with varying difficulties by implementing **Minimax** search with alpha-beta pruning
- Reduced the search runtime by **over 70 seconds** per turn using pruning methods, heuristics and optimizations

Mole-croscope :: CuHacking 2nd Place, Innovapost Challenge Winner

Feb 2019

- Collaborated with a team of 4 to build a mole-tracking **Android** app with skin cancer prediction trained on **GCP**
- Designed mole anomaly-tracking through object translation invariance with **Python** and **OpenCV**
- Implemented **Python** preprocessing scripts to auto-correct skew and translation differences between user inputs

>> EXPERIENCE

Contest Organizer and Moderator :: DMOJ: Modern Online Judge

July 2020 - Present

- Moderated Canada's largest online judge, dmoj.ca, with over **100 000 users**, 300 contests, and 3 500 problems
- Co-authored and organized **7** programming contests averaging **200+** participants each
- Devised **graph theory**, **data structures** and **dynamic programming** problems and published solution editorials
- Ensured fair point allocation across tier levels by implementing solutions approaches and preparing test cases

>> AWARDS

- Canadian Computing Olympiad (CCO 2021) :: **Silver Medalist**
- Canadian Computing Competition (CCC 2021) :: **Perfect Score, top 1.1%** (~3 000 participants)
- Facebook Hacker Cup :: **Top 1.2%** in 2020, **top 1.6%** in 2021 (~35 000 participants)
- Google Kickstart Competition 2021 :: Round C **top 0.4%**, 48th worldwide (~12 200 participants)
- Google Kickstart Competition 2020 :: Round G **top 0.9%**, 73rd worldwide (~8 100 participants)
- Advent of Code :: **Top 0.2%** in 2019, **top 0.03%** in 2020, **top 0.017%** in 2021 (~210 000 participants)
- USA Computing Olympiad 2019, 2020, 2021 :: **Platinum division**

>> EDUCATION

University of Waterloo :: B. of Software Engineering

Sept 2021 - Present

Candidate for Bachelor of Software Engineering, Honours, Co-operative Program

Expected graduation April 2026

GPA: 3.93 / 4